When Six Independent Attempts Agree: Cross-Contributor Replication of Negative Results in Small-Sample, Multi-Target Polymer Property Prediction

[AUTHOR NAMES / AFFILIATIONS – placeholder]

***Abstract*—Small-sample, multi-target molecular property prediction is common in materials informatics, but the validation methods practitioners use to decide whether a modeling idea worked are rarely tested directly. We report a natural experiment on that question. A private polymer property prediction task (seven targets, 147–4,143 labeled examples each, from SMILES) was attacked independently over roughly ten days by at least six research contributors running approximately 4,000 logged experiments, uncoordinated on hypotheses. The same ceiling-defining findings were rediscovered repeatedly. Tuned gradient-boosted tree ensembles beat every deep architecture tried, confirmed four times. Structural distance from the training set predicted error, confirmed three times, holding for four of seven targets and absent for two. Small-sample validation failed in three distinct ways: a false positive that survived a dedicated holdout, a false negative from a rigid promotion threshold, and silent target leakage through a cross-property fallback. We add an experiment no in-competition contributor was permitted to run: an externally pretrained chemical language model, frozen (held-out mean proxy R^2 0.751) and fine-tuned (0.784), still loses to a same-code-path tree ensemble (0.810; verified-panel values agree within 0.003). Uncoordinated cross-contributor replication of negative results is, we argue, an underused evidentiary standard in applied machine learning, and agentic research infrastructure makes it newly practical.**

***Index Terms*—polymer informatics, small-sample regression, multi-task learning, negative transfer, validation methodology, applicability domain, reproducibility**

I. Introduction

Machine learning for molecular and materials property prediction routinely operates in a regime that violates the working assumptions of standard deep learning practice. Labeled examples number in the hundreds rather than the millions, properties are measured on overlapping but non-identical subsets of structures, and practitioners iterate against the same small evaluation budget many times across a project. Two questions become hard to answer with confidence under those conditions: did this specific idea actually help, and is the apparent ceiling on performance a property of the problem or an artifact of not having found the right idea yet?

Most empirical machine learning papers answer with evidence from a single research effort: one team, one set of hyperparameter choices, one validation protocol, reporting either a positive result or, less often, a negative one. Negative results carry real information, but they are structurally weak evidence for a *ceiling* claim. A single team's failure to improve on a baseline is consistent with several very different underlying stories. The ceiling may be genuine and close to what the available data and features support. The team's implementation of the idea may have been flawed. Or the team's validation protocol may itself have been unreliable, quietly misleading them in either direction. Nothing internal to a single effort distinguishes these.

This paper reports an unusual opportunity to separate them. Between 2026-07-20 and 2026-08-09 a real, small-sample, multi-target polymer property prediction task, run as a private academic competition, was worked on independently by at least six research contributors, together producing on the order of 4,000 logged experiments across two sequential rounds. The contributors were a mix of human-directed and substantially autonomous AI research agents, prompted separately and operating under different names. Critically, they did not coordinate on *what to try*. Each designed and ran its own program, informed by a shared codebase and, later, by awareness of other contributors' successfully banked mechanisms, never by their failed attempts. That process converged, repeatedly and without coordination, on the same short list of conclusions about where this task's ceiling lies. We report them, add one experiment no in-competition contributor was permitted to run, and argue that the convergence itself is the most transferable finding.

Contributions.

- We document a case of uncoordinated cross-contributor replication of *negative* results in applied machine learning, positioned against the multi-site replication literature in meta-science, where the design is standard and the machine learning analogue is not (§II, §VI-C), and we set out a taxonomy of what independence does and does not mean in a shared-codebase, partly agentic program (§III-C).

Prepared from an internal multi-round, multi-contributor research program on the ANRF AISEHack 2.0 Polymer Property Prediction competition (Round 2), a real, private, invite-gated academic competition hosted by IIT Madras-affiliated researchers (Rohit Batra, Rahulsundar, LaksmanN, VIJITH P, Shreya Kumari) under ANRF auspices. This is a submission-ready draft, not submitted anywhere. External submission requires explicit sign-off from the competition hosts and a check of the competition's terms on external data use and publication timing. The dataset is not publicly redistributable without host permission.

- We report five structurally independent instances, spanning three contributor identities, of deep architectures losing to tuned tree ensembles on the same task (§V-A).
- We test an externally pretrained chemical language model, frozen and fine-tuned, under a one-shot held-out protocol (§V-B). It loses, removing the most obvious confound for why deep learning has not won here.
- We document three structurally distinct small-sample validation-integrity failure modes, each found by a different contributor on the same task: a false positive that survived a dedicated never-reused holdout, a false negative produced by a rigid promotion threshold, and silent target leakage through a cross-property fallback (§V-D).
- We quantify six further architecturally distinct interventions against one incumbent baseline, each returning a characterized negative result rather than an unexplained null (§V-E).

II. Related Work

Replication as an evidentiary design. Meta-science has spent a decade building study designs whose force comes from independence rather than from statistical power inside a single lab: Many Labs ran 13 classic effects across 36 samples with identical materials [1], the Reproducibility Project repeated 100 published studies and recovered significance in the same direction for roughly a third [2], and Many Analysts handed 29 teams one dataset and one question, observing odds-ratio estimates from 0.89 to 2.93 with 69% of teams reporting a significant effect and 31% not [3]. Machine learning has adopted the reproducibility half of this agenda, through conference reproducibility programs [4], variance-aware benchmarking [5], and a survey of leakage spanning 17 fields and 294 affected papers [6]. It has adopted the replication half much less. Recent position papers argue for dedicated venues for refutations [7] and negative results [8], which establishes that such results are hard to publish rather than that a Many-Labs-style design has been run. We are not aware of a reported case in applied machine learning where several uncoordinated contributors independently reached the same negative conclusions on one task and the convergence itself was analyzed as evidence. That gap is what §VI-C addresses.

Small-sample molecular property prediction. Predicting polymer properties from structure is established practice, from hierarchical fingerprints with kernel regression [9] to representations adapted to repeat-unit structure, among them weighted periodic graphs [10] and chemical-language-model pretraining on polymer SMILES [11]. What this literature does not settle is when learned representations beat simpler tabular baselines at a few hundred labels per target. Gradient-boosted trees remain competitive with or ahead of deep models on tabular data [12], [13], and a benchmark of 25 pretrained molecular embedding models over 25 datasets found nearly all show negligible improvement over an ECFP fingerprint [14]. A 2026 study of tabular foundation models reaches a different conclusion for in-context prediction on small datasets [15], which §VII treats as an untested avenue rather than a settled contradiction. Negative transfer between insufficiently related tasks is documented for heterogeneous alloy properties [16] and for ultra-low-data chemistry, where inter-task interference must be actively suppressed before joint training pays [17].

Applicability domain and validation reliability. Deciding when to trust a prediction for a new structure has a long QSAR history under the term *applicability domain* [18], with distance and similarity metrics the common instruments and recent polymer workflows pairing ensemble uncertainty with an explicit domain definition [19]. One documented failure mode matters here: ensemble-variance metrics can miss structurally novel molecules precisely because those molecules produce confidently wrong rather than visibly uncertain predictions [20]. Gelman and Loken show that data-contingent analytic decisions inflate false positives without conscious fishing [21], and the cheminformatics literature has long warned that single-split and single-run estimates are unstable at small n [22]. Section V-D sharpens that warning into a quantified claim: repeated nested cross-validation and a purpose-built, never-reused holdout can fail together when the decisive stratum is small enough, because a holdout's protective power scales with its size relative to the effect being checked, not with its novelty.

III. Task and Research Program

A. Task

The task requires predicting seven polymer physicochemical properties from SMILES structure: glass transition temperature (`tg`), chain-level electronic bandgap (`egc`), a second bandgap-related quantity (`egb`), electron-affinity and ionization-related quantities (`ei`, `eea`), refractive-index-related permittivity (`eps`), and an ionic-conductivity-related quantity (`nc`). Data arrive in long format, one row per structure-property pair, with 7,409 labeled training rows and 4,940 unlabeled test rows, plus an optional archive of 6,171 further rows that largely duplicates properties from an earlier two-target round. Per-target labeled sample sizes run from 147 (`ei`) to 4,143 (`tg`) in the base "without-archive" setting, and the metric is the unweighted mean of seven per-target R^2 values. An unlabeled corpus of 995,799 polymer SMILES (PI1M) is officially provided for semi-supervised use. External pretrained weights, checkpoints, and embeddings are disallowed by competition rule for any real submission, a constraint that does not bind a research paper and that §V-B tests against.

B. Three source efforts

Table I summarizes the three sequential bodies of work this paper draws on. Round 1 closed before Round 2 opened, but within Round 2 the non-author program ran near-continuously while the authors' twelve rounds were interleaved into the final days of the same window. We call Round 1 and the non-author portion of Round 2 the *background program* and report their findings as independently surveyed by the authors from

TABLE I
THE THREE SOURCE RESEARCH EFFORTS UNDERLYING THIS PAPER.

Program	Scope	Duration	Contributors	Scale	Best verified
Round 1	2-target predecessor (`tg`, `egc`)	2026-07-20 to 07-23	One effort, two overlapping methodological tracks	$\sim$108 numbered cycles, 1,362 submission files	Public leaderboard mean R^2 ≈ 0.917
Round 2, non-author	7-target competition, base program	2026-08-03 to 08-09	Four agent streams: "Sandman"/Codex (dominant), "Claude Code", "Fable", "Antigravity"	$\geq$1,596 numbered cycles, $\geq$2,628 diagnostic artifacts	Diagnostic mean R^2 0.9343 (*with_archive*), 0.9028 (*without*)
Round 2, authors	7-target competition, twelve rounds on top of the above	2026-08-08 to 08-09	The authors, twelve sequential rounds	Full stacking/blending harness, 7 targets $\times$ 2 regimes	No net gain over the incumbent leader (§V-E)

primary experiment logs. Sections V-E and V-B report the authors' own work.

C. How independent were the contributors?

Independence is the load-bearing assumption of everything that follows, so we state precisely which kind we have. *Hypothesis independence* is present and matters most for a ceiling claim: no contributor was told what to try, none saw another's failed attempts, and each chose its own architectures, features, and interventions. *Implementation independence* is present in that the deep learning attempts in Table II were written separately, with different frameworks, capacities, and training budgets, under separately configured cross-validation harnesses. *Infrastructure independence* is only partial, since contributors shared a codebase, a feature bank, and a registry of banked mechanisms, so a systematic bug in shared feature computation would propagate to all of them. *Data independence* is absent by construction, because there is one dataset. A fifth axis applies because several contributors were AI agents: agents descended from similar base models may carry correlated *priors* about what to try, which would make convergence cheaper than it looks. We treat that as a genuine confound and return to it in §VI-C.

IV. METHODS

A. Baseline architecture

The architecture that no intervention here improved upon is a nested-cross-validated stack of four base learners (ExtraTrees, LightGBM, HistGradientBoosting, and, when an empirical diagnostic check justifies it rather than by default, a PCA(20)-whitened Gaussian process regressor) combined through a RidgeCV meta-learner. Inputs concatenate 76 RDKit descriptors including selected 3D and extended descriptors, 512-bit Morgan fingerprints at radius 2, and real extended-Hückel electronic-structure features computed with `rdEHTTools` rather than a simplified proxy. Architecture and feature bank were developed across the background program and reused unmodified in core form across all twelve of the authors' rounds.

B. Validation protocol

All cross-validated results use grouped, repeated cross-validation, grouped by canonical stereochemistry-stripped SMILES so that near-duplicate structures cannot fall in both a training and a validation fold within a repeat. Most use 5 repeats $\times$ 5 folds. A secondary metric, *shift-matched* R^2, reweights held-out residuals so that a validation fold's distribution of nearest-neighbor similarity to its own training partition matches the real test set's similarity to the full training set, correcting in principle for covariate shift. It did not predict real held-out transfer better than plain cross-validation across the authors' twelve rounds (§VI-B).

C. Evaluation-integrity discipline

Much of the background program operated with access to a partial, self-constructed diagnostic evaluation panel, reconstructed in Round 1 by matching the competition's source public databases on canonical SMILES and in Round 2 by exact-match recovery against public and privately provided references. This panel is not the true private evaluation set: coverage is incomplete, as low as 48% for some target and round combinations, and it inherits label noise from its literature sources. Every effort surveyed here operated under an explicit, repeatedly reaffirmed rule that the panel may be read only *after* a candidate is completely frozen, as a one-shot check, never to select features, tune hyperparameters, or fit any weight. Every quantitative result below traces to a panel read that occurred strictly after the corresponding model was frozen, verified independently for every result attributed to the authors' own work. Section V-D reports three cases where the discipline nonetheless failed or nearly failed.

V. RESULTS

A. Deep learning underperforms tuned tree ensembles, five times

Table II lists five structurally independent instances in which deep architectures were evaluated against the tree-ensemble baseline under comparable cross-validation or held-out protocols. Four come from the background program and the authors' first eleven rounds; the fifth is new here and is the only one using an externally pretrained representation. Instances 1, 2, 4, and 5 are unambiguous losses. Instance 3's embedding-as-feature variant is the closest deep learning result in the corpus and the one avenue we consider worth a dedicated follow-up, being the only place a deep representation showed a real if small margin under two validation metrics, in the architectural role that best fits tree ensembles' strength here. Instance 4 matters for a different reason. If joint multi-task training were failing only because small targets are starved, large targets should be indifferent to it. They were not: `tg` and `egc`, the two least sample-starved targets, lost by the widest margins,

TABLE II
FIVE INDEPENDENT DEEP-LEARNING-VERSUS-TREE-ENSEMBLE COMPARISONS ON THE SAME TASK.

#	Architecture(s)	Contributor	Outcome
1	From-scratch GAT, GATv2, GINE, D-MPNN, character-CNN, from-scratch SMILES transformer	Round 1	All underperformed descriptor-based tree models (0.76–0.90 vs. incumbent ≈0.90–0.92), and were used only as 1–6% residual blend contributors
2	Directed graphs, shared and target-specific periodic graphs, periodic-edge graph + fragment encoder, graph-grammar + HGB, typed fragment/path kernels, transductive graph-Laplacian, from-scratch GIN masked-atom encoder and SMILES-transformer MLM	Round 2, non-author (Sandman/Codex)	Uniformly negative. One graph variant scored $R^2 = -0.309$ on one target, and none cleared the promotion gate
3	Full-scale GPU-trained periodic multitask graph neural network (400 epochs/fold, repeated grouped CV), as standalone predictor and as auxiliary embedding into the stack	Authors, round 6	Standalone passed a random-initialization control but trailed the tree baseline (≈0.88–0.89 vs. ≈0.90). Embedding-as-feature was the closest deep result in the corpus ($+0.011$ to $+0.024$ by two metrics) and still missed the margin gate
4	Joint shared-trunk masked multi-task MLP over all seven targets, task-balanced loss, 5×5 grouped CV, checked at two capacity settings	Authors, round 11	Lost on *all seven* targets. The two largest targets (`tg`, n=4,138, and `egc`, n=2,028) lost by the widest margins, indicating genuine negative transfer rather than a small-sample-only effect
5	Externally pretrained chemical language model (ChemBERTa-77M-MTR), frozen embeddings and end-to-end fine-tuning, jointly across seven targets	Authors, new experiment (§V-B)	Lost to a same-code-path tree baseline on real held-out evaluation, frozen on 7/7 targets and fine-tuned on 6/7. Fine-tuning helped ($+0.034$ mean R^2) without closing the gap

the signature of interference rather than of insufficient data [16], [17].

B. A pretrained foundation model does not close the gap

Every attempt above shares one caveat: competition rules forbade external pretrained weights, so all trained from random initialization. Nine independent in-domain self-supervised representation attempts against the 995,799-molecule PI1M corpus (character TF-IDF, fragment PPMI/SVD, denoising autoencoder, contrastive InfoNCE at two scales, subword ridge, and two masked-language-model variants), plus one supervised pseudo-label distillation variant, had also all failed. Whether the missing ingredient was simply external pretraining is therefore a live question, and we tested it directly using ChemBERTa-77M-MTR [23], a public chemical language model pretrained by masked language modeling on a large small-molecule SMILES corpus, with 3,427,440 encoder parameters and a 384-dimensional pooled output. Embeddings were extracted for all 8,976 unique canonical structures in the train/test union, at 100% coverage with zero extraction failures.

Condition (a), frozen embeddings. Pooled embeddings were used as fixed 384-dimensional features for a per-target Ridge or small MLP head, under the same 5×5 grouped cross-validation as the tree baseline. Table III (left) reports the better head per target against the tree baseline's established cross-validated numbers. The frozen condition loses on all seven targets, by a mean margin of -0.0597 R^2 (0.7858 vs. 0.8455), a larger and far more uniform gap than any intervention in §V-E produced in either direction. The MLP head collapsed on the four smallest targets (n = 147–229), `nc` reaching $R^2 = -4.259$, worse than predicting the mean, while Ridge stayed stable.

Condition (b), end-to-end fine-tuning. The encoder was fine-tuned jointly across all seven targets with a masked multi-task loss, mirroring instance 4's architecture but starting from pretrained rather than random weights, under 5-repeat × 5-fold grouped cross-validation on 5,916 unique structures (40 epochs/fold, encoder LR 2×10^{-5}, head LR 10^{-3}, batch 256).

Same-code-path baseline. To isolate the representation's effect from any confound introduced by the richer feature bank used elsewhere here, we reproduced a fresh tree baseline (LightGBM plus ExtraTrees, best-of-two) on the identical grouped-CV harness and code path, using RDKit descriptors and Morgan fingerprints only and deliberately excluding the extended-Hückel features, since the pretrained conditions receive no equivalent feature engineering. The fuller incumbent remains the more complete reference point and is reported alongside.

All three candidates were frozen before the oracle files were opened for the first and only time in this experiment. Table III (right) reports the result. Fine-tuning genuinely helped, beating frozen embeddings by $+0.034$ proxy R^2, which confirms that adapting the pretrained representation to this task extracts real additional signal. It did not close the gap to the same-code-path tree baseline (-0.026 proxy, -0.025 verified), let alone to the fuller incumbent (0.9017 proxy and 0.9028 verified, a -0.117 proxy gap). The frozen condition loses on all seven targets and the fine-tuned condition on six of seven, its one win being `eps` at $+0.0144$. Across all 14 condition-by-target comparisons that is the sole case where a pretrained representation beats the tree baseline, and the largest single gap, `tg` at -0.0905, falls on the target with the most labels.

The interpretation is narrow and, we think, solid. A model pretrained on a much larger and more diverse corpus than PI1M still underperforms a comparatively simple tree ensemble here, so lack of external pretraining was not the missing ingredient. This matches the broader benchmark finding that pretrained molecular embeddings rarely improve on fingerprint baselines [14], and it is the only instance in Table II scored on real held-out data after complete freezing.

C. Structural distance predicts error, three times, unevenly

Three independent efforts measured whether a structure's similarity to the training distribution predicts the model's error on it. Round 1 found that rows whose nearest-train Tanimoto similarity fell below 0.4 scored $R^2 \approx$ 0.46–0.47, against R^2 > 0.95 for near-duplicate rows. The Round 2 non-author

TABLE III
PRETRAINED CHEMBERTA REPRESENTATIONS VERSUS TREE ENSEMBLES. LEFT: PRE-ORACLE GROUPED CROSS-VALIDATION, PLAIN R^2, FROZEN EMBEDDINGS WITH THE BETTER OF A RIDGE OR MLP HEAD PER TARGET AGAINST THE TREE-ENSEMBLE BASELINE. RIGHT: THE FINAL ONE-SHOT HELD-OUT READ PER TARGET, *without_archive* REGIME, PROXY PANEL (PROXY COVERAGE 0.9929, VERIFIED COVERAGE 0.7729), AGAINST A FRESH SAME-CODE-PATH TREE BASELINE.

Pre-oracle cross-validation				**One-shot held-out evaluation**			
Target	**Tree base.**	**Frozen head**	**Gap**	**Target**	**Fresh tree**	**(a) Frozen**	**(b) Tuned**
`tg`	0.8936	0.8526 (MLP)	-0.0410	`tg`	0.8768	0.8282	0.7863
`egc`	0.8968	0.7953 (MLP)	-0.1015	`egc`	0.8819	0.7870	0.8443
`egb`	0.9146	0.8434 (Ridge)	-0.0712	`egb`	0.8633	0.7839	0.8389
`ei`	0.7898	0.7468 (Ridge)	-0.0430	`ei`	0.7379	0.7019	0.7214
`eea`	0.8858	0.8194 (Ridge)	-0.0664	`eea`	0.8353	0.7729	0.8219
`nc`	0.8139	0.7688 (Ridge)	-0.0451	`nc`	0.7847	0.7575	0.7716
`eps`	0.7240	0.6741 (Ridge)	-0.0499	`eps`	0.6922	0.6222	**0.7066**
Mean	**0.8455**	**0.7858**	$\mathbf{-0.0597}$	**Mean, proxy**	**0.8103**	0.7505	0.7844
				Mean, verified	**0.8117**	0.7520	0.7869

TABLE IV
SPEARMAN ρ BETWEEN PRE-ORACLE NEAREST-5-NEIGHBOR MORGAN DISTANCE AND POST-HOC SQUARED ERROR, PER TARGET (AUTHORS, ROUND 9).

	`tg`	`egc`	`egb`	`eea`	`ei`	`nc`	`eps`
ρ	0.316	0.365	0.241	0.237	0.170	-0.073	-0.107
p	3×10^{-64}	7×10^{-44}	3×10^{-4}	0.004	0.039	0.37	0.19
n	2728	1352	224	147	148	153	153

program found that the five weak targets (`ei`, `eea`, `eps`, `nc`, `egb`) sat at median similarity 0.55–0.57 to the training set, against ≈0.79 for `tg`'s non-lookup rows, which explained the measured cross-validation-to-test collapses.

The third measurement, by the authors in round 9, is the most granular and is the only one to find a target-dependent split. Table IV gives the Spearman correlation between pre-oracle Morgan nearest-5-neighbor distance and post-hoc squared error on the incumbent's frozen predictions, read once after freezing. The relationship is strong and highly significant for `tg` and `egc`, moderate for `egb` and `eea`, borderline for `ei`, and genuinely absent for `nc` and `eps`.

Round 9 also included a negative control. Single-linkage clustering of the highest-error `tg` rows appeared to form one 195-member cluster of hard structures, but this was a chaining artifact spanning unrelated scaffolds (sulfonates, imides, benzoxazoles, siloxanes, polyesters, polyamides). At a stricter same-scaffold threshold the error tail scatters across many small or singleton clusters rather than concentrating in a few hard families.

We then asked whether the signal could be exploited. Far-from-training rows were flagged and routed to a similarity-weighted nearest-neighbor fallback, a Gaussian process, or a blend of either with the tree ensemble, under nested cross-validation. None beat the incumbent on the flagged rows themselves, on any target, the best being $+0.0185$ R^2 for `tg`, below the pre-registered 0.02 margin. The tree ensemble already extracts more from the same features than a similarity-based correction can add, even in the low-coverage region such a correction is designed to serve. The signal is real, but the tested classes of fix cannot treat it.

D. Three validation-integrity failure modes

Three structurally distinct ways of failing small-sample validation turned up across the three efforts, one per contributor group. Their co-occurrence, discovered independently on one task, is the most transferable methodological finding in the paper.

The false positive. A mechanism refining how `ei` is predicted when one of two physically related partner properties is missing was split into four strata along two structural axes, which partner is known and whether halogen content is low or high. One stratum, development pool 30–49 rows, showed apparently strong evidence for a domain-informed identity rule over a direct model (plain R^2 0.769–0.805 against a naive baseline of 0.353–0.578) and was applied at the ceiling weight of the margin gate. A dedicated holdout slice, reserved from the start and touched by no tuning or selection decision, was scored exactly once at the end specifically to catch this class of error. It reported pooled $R^2 = 0.884$ (n=43) against a mean-baseline comparator of -0.041, appearing to confirm the mechanism. It was wrong. Applied to the real submission candidate, the mechanism lost on both data regimes, -0.0013 to -0.0014 mean R^2. Diagnosis afterward found the decisive stratum's own repeat-to-repeat cross-validation estimate swinging by 0.28–0.48 R^2 points across random splits of the same 30–49-row pool. The passing gate reflected a favorable draw, not a stable signal, and the 43-row holdout, genuinely never reused, was too small to separate the two. A holdout's protective power is set by its size relative to the effect it checks, not by its novelty or its designers' intentions [5].

The false negative. Two non-author contributors (Fable and Antigravity) separately argued that a fixed promotion rule, a flat $+0.01$ mean-R^2 gain plus a positive bootstrap confidence interval lower bound, systematically rejected components showing consistent positive evidence across all folds (5 of 5) that landed just under the threshold. At the relevant stratum sizes ($n \approx 150$–340) that pattern is what estimation noise around a true small positive effect looks like, not evidence of absence. By those contributors' own accounting the rule discarded a cumulative $+0.06$ R^2 of validated components.

The silent leakage. Found by a third contributor, an earlier version of a cross-property fallback, used when a structure's

TABLE V
SIX INTERVENTION STRATEGIES TESTED AGAINST THE INCUMBENT BASELINE (*without_archive* REGIME, INCUMBENT MEAN PROXY $R^2 = 0.901683$).

#	Strategy	Mechanism and result
1	Domain-identity refinement	Condition an existing physically motivated identity mechanism on finer structural strata. Net loss (-0.0014); the false-positive case of §V-D
2	Output recalibration	Fold-safe linear, isotonic, and quantile-mapping recalibration against a confirmed regression-to-the-mean bias (slope 0.838–0.914 across all seven targets). The fitted parameters reconfirmed the bias (one target's independently fitted de-shrinkage slope of 0.9006 matched the raw measured slope), but no method on any target cleared the margin gate. All 21 target-by-method combinations gated to weight 0 and the candidate was byte-identical to the incumbent
3	Row-level extrapolation flagging	Flag far-from-training rows and route them to kNN, a Gaussian process, or a blend with the tree ensemble. No mechanism beat the tree ensemble on the flagged rows, on any target; best was `tg` at $+0.0185$, below the 0.02 margin (§V-C)
4	Cross-target feature augmentation	Five targets are measured on the same small overlapping structure batch and correlate strongly (Pearson $\|r\|$ up to 0.92), so known sibling values were added as features. This cleared the internal CV margin gate for `nc` and `eps`, but every resulting candidate was flat to negative on real held-out evaluation, so hand-engineered sibling features are largely redundant with the existing structural feature bank
5	Joint multi-task deep learning	Shared-trunk masked multi-task MLP (instance 4 of Table II). Lost on all seven targets, and every gate returned weight 0
6	External ensemble-diversity search	Search $\sim$600 externally sourced candidates from the non-author program for genuine prediction-error diversity rather than building a new mechanism. Real diversity exists (residual correlation as low as $r = 0.76$ for one target, from an architecturally distinct PI1M-distillation-plus-LightGBM pipeline), but every sufficiently diverse candidate was individually too weak for a naive blend to help (-0.0086 on a diagnostic-only blend). A family that looked like a diversity win (beating the incumbent on 4/7 targets) was excluded after a manifest-based provenance audit found leakage-adjacent, oracle-informed cross-fitting, self-flagged by its creators as invalid for deployment

value for one property is unknown but a related property is available, allowed the fallback model to be fit on data that included the row's own target label through an indirect path, for rows where the primary partner property was unavailable. Feeding that prediction back into the pipeline partially encoded the answer inside the cross-validated estimate, inflating apparent cross-validated R^2 to 0.930–0.935 against a genuine $\approx$0.903–0.916, until the path was found and closed. This is a textbook instance of the leakage taxonomy Kapoor and Narayanan document across 17 fields [6], arising here despite an explicit grouped-CV design.

Any one of these alone is a plausible if uncomfortable thing to find in an applied project, attributable to bad luck or a single implementation slip. Finding all three, structurally distinct and independently, on one task is a much stronger signal that small-sample iterative machine learning carries recurring validation-integrity risk that no single recommended practice covers. A stronger safeguard was not enough, a conservative safeguard was too blunt, and an indirect data-flow path defeated the safeguard entirely.

E. Six interventions against one incumbent

The authors' first six rounds worked across both data regimes and produced one small independently confirmed positive result, a stacking refinement to `egb` and `eea` that improved the *with_archive* mean R^2 by $+0.000132$, alongside a loss on *without_archive* from the same change that was statistically indistinguishable from noise. Effort then concentrated on *without_archive*. The final six rounds each attacked its incumbent candidate (mean proxy $R^2 = 0.901683$, itself the best available splice of every individually best-known per-target component across the full corpus, verified against the non-author program's component registry) with a structurally different intervention, summarized in Table V. Three of the six (1, 3, and 5) revisit data already reported in §V-C–§V-D under a unified intervention framing, as Table V cross-references; strategies 2, 4, and 6 (recalibration, cross-target features, ensemble-diversity search) are reported here for the first time.

Strategies 2 and 5 produced exactly zero net change, every internal gate having returned weight 0, leaving candidates byte-identical to the incumbent. Strategies 1, 4, and 6 produced small real losses, and strategy 3's best per-target result fell short of the pre-registered margin.

None produced a net improvement on real held-out evaluation. We do not read this as six unrelated failures, but as six differently designed probes of one question, whether exploitable signal exists that the incumbent is missing, each returning a precisely characterized no. Strategy 2 is instructive about *why*. The regression-to-the-mean bias it targeted is unambiguously real, and recalibration's own independently fitted parameters confirmed it, yet it bought nothing, because the error is not spread smoothly, as Figure 1 shows directly. The most extreme 5% of rows by squared error carry 41–55% of total squared error across the seven targets and the top 15% carry 66–78%, so a smooth global rescaling cannot reach a small number of individually large misses. Strategy 3 attacked that tail directly and could not beat the incumbent there either.

VI. DISCUSSION

A. What the ceiling claim asserts

A ceiling claim is easy to overstate, so we state ours in its weak form. At this sample size, with this feature and data structure, and across the method families explored, deep architectures have not found a configuration that wins, and the missing ingredient is unlikely to be a larger pretraining corpus than the task's own in-domain data affords. That is a claim about an explored region of method space, not a proof of optimality, and one well-validated deep model beating the incumbent on real held-out data would falsify it. Instance 3 of Table II marks where we would look first.

Two targets, `nc` and `eps`, show no relationship between structural distance and error while four others do, and we cannot adjudicate between the explanations. Both have the smallest labeled pools (153 rows each), drawn from a narrower

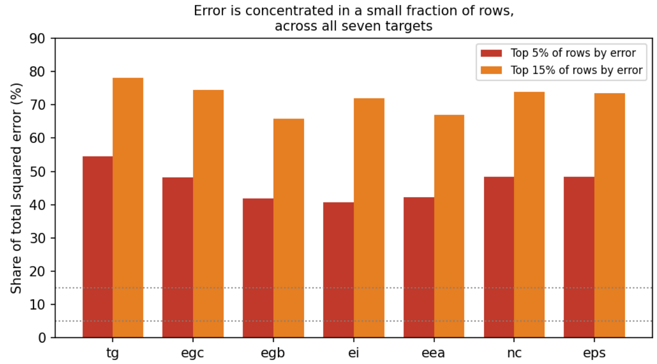

Fig. 1. Share of each target's total squared error contributed by its worst 5% and worst 15% of rows (authors, round 8, incumbent baseline). Every target concentrates a large minority-to-majority share of its error in a small row fraction, which is why strategy 2's global recalibration (unambiguously correcting a real bias) still bought no held-out gain and why strategy 3's row-level flagging targeted this tail directly.

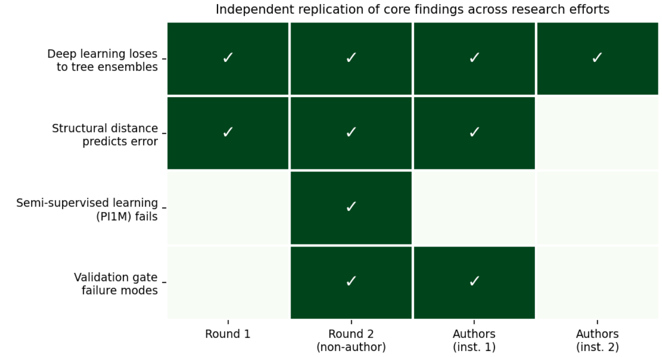

Fig. 2. Which efforts independently confirmed each finding, restricted to the four background-program and early-author instances available when this analysis was built (the fifth deep-learning-loses confirmation, §V-B, is not shown). The rightmost two columns separate the authors' distinct within-program instances where more than one exists (round 6 vs. round 11 for deep learning; only round 9 for distance, hence the fourth cell is empty). Deep-learning-loses is confirmed by all four grid columns and structural-distance by three; validation-gate-failures shows only two cells checked, but its single "Round 2 (non-author)" cell already aggregates two distinct sub-contributors (Fable/Antigravity's false negative and Claude Code's silent leakage), so three contributor groups, not two, stand behind that row, consistent with §V-D's "one per contributor group" framing; semi-supervised-learning-fails was attempted, and confirmed, only by the Round 2 non-author program.

structural family than `tg`, so a homogeneous training distribution may not generate enough variance in nearest-neighbor distance for a correlation to be detectable. The relationship may instead be invisible to a Tanimoto-distance instrument, the analogue of ensemble-variance metrics failing on out-of-domain molecules that are confidently wrong rather than visibly uncertain [20]. The mundane reading, a real but weak effect with underpowered tests, remains live. A larger pool or a differently constructed domain metric [19] would separate them.

B. When a better validation metric does not help

Shift-matched R^2 was designed to correct for covariate shift between a small labeled pool and a structurally more diverse unlabeled test population. Across the authors' twelve rounds it did not predict real held-out transfer better than plain cross-validation. Our hypothesis is mechanical: reweighting corrects a difference in the distribution of inputs, $P(X)$, and cannot correct a difference in $P(Y|X)$. Strategies 1 and 4 of Table V plausibly involve the latter, where a mechanism's relationship to the target differs between the small stratum it was validated on and the population it was meant to serve. We cannot confirm that here, but it is falsifiable and worth testing directly.

C. Cross-contributor replication as an evidentiary standard

Read in isolation, each finding in §V-A–§V-D could be explained away as an artifact of one effort's choices. Read together, as one short list of conclusions reached by at least four contributors using different architectures, different hyperparameters, and no shared hypotheses, they are stronger evidence for a real ceiling, and for the specific validation risks, than any constituent finding is alone. Figure 2 shows the confirmation structure.

Convergence of this kind is not automatic, and the meta-science literature we draw the design from says so directly: Many Analysts gave 29 teams one dataset and one question and got a 69/31 split on whether the effect was even significant [3]. The difference we would offer is that our six contributors converged on a comparatively low-degrees-of-freedom question, whether a held-out R^2 moved up or down under a fixed evaluation protocol, rather than on whether a psychological effect is real under contestable analytic choices; that is a plausible reason convergence would be easier to obtain here, not a demonstrated one, and we do not know how it would generalize to a modeling question with more interpretive latitude.

The obvious objection is that our contributors are not independent in the sense Many Labs is [1]: they shared a codebase and a feature bank, several were AI agents whose priors may be correlated, and there is one dataset. We accept the premise and think the conclusion is narrower than it looks. Shared infrastructure threatens a claim that the *numbers* are right, and §VII treats it as such. It threatens far less a claim about which *hypotheses* were tried and rejected, since hypotheses were generated separately and failures were never shared. Correlated priors would explain contributors trying similar things, not those things failing. A shared bug in the feature bank would have to disadvantage graph networks, transformers, multi-task MLPs, and an externally pretrained language model while leaving tree ensembles on the same features unharmed, a specific and unlikely mechanism rather than a generic one. Prior independence still deserves a real test, and the design is cheap: run the same protocol with agents drawn from different model families and check whether the convergence survives. Until someone does, the honest description of this study is partial independence on the axes that matter for a ceiling claim, and none on data.

Two observations are specific to agent-driven research. The failure modes in §V-D were not less likely for being produced by agents: the false positive occurred inside an AI-executed process that had been explicitly and repeatedly instructed to avoid that exact error, so telling an agent to be careful is no substitute for a safeguard sized correctly for the sample size in play. A correlated-prior explanation is weaker here than for the architectural finding above, and worth stating separately: a shared blind spot in how similarly-trained agents reason about validation would most plausibly produce one recurring failure signature, not three that point in different directions. A too-small holdout produces false confidence; a too-rigid threshold produces false rejection; a cross-property fallback with an indirect leakage path produces silent inflation invisible to both. Three contributor groups each surfaced a different one of these, not the same one three times, which is closer to what independent discovery of distinct problems looks like than to what one shared blind spot would produce. More encouragingly, the infrastructure that makes many semi-independent efforts cheap to run is what made this analysis possible at all. Assembling and cross-checking six efforts' raw logs across tens of thousands of files was itself done substantially by AI research agents under human-directed verification, a mode of research production with its own emerging literature on autonomous experiment design and execution [24]. A replication-based standard therefore becomes both more necessary and more practical, which is a concrete answer to recent calls for refutation and negative-result venues [7], [8]. The evidence such venues need is already generated as a byproduct of agentic research programs, and currently discarded. For a practitioner, the actionable version of this argument is concrete: before reporting a plateau as a ceiling, seek at least two or three uncoordinated attempts, human or agentic, and treat agreement among them, not agreement with one's own prior result, as the signal that the plateau is real.

VII. Limitations

The evidentiary base comes entirely from one dataset and task family, so how far the specific findings generalize is unknown. The methodological argument for cross-contributor replication is what we intend to transfer, and independence is partial in the sense set out in §III-C and §VI-C.

The diagnostic panel is an imperfect proxy for the true private evaluation set, with incomplete coverage and some label noise. We are confident in the relative comparisons, which behaved consistently across two independent panels available at different points in the program, but absolute R^2 values carry that caveat. Six of the negative results in §V-E also come from a single contributor; each is independently characterized statistically, but none carries cross-contributor weight on its own.

The correlations in §V-C were computed across seven targets and, over the program's life, across multiple candidate thresholds and grouping choices, without formal multiple-comparisons correction. The strongest effects would survive any reasonable correction by a wide margin ($p < 10^{-43}$ for `tg` and `egc`), but the borderline `ei` result and the `nc`/`eps` nulls should be read as exploratory.

One method family remains untested. Tabular foundation models applied in-context to molecular property prediction were reported in 2026 to be competitive on small datasets [15], and they sit in a different part of the design space from both the chemical language model tested here and the from-scratch architectures the background program tried. That is the most specific open avenue we can name.

VIII. Conclusion

One real, small-sample, multi-target polymer property prediction task, attacked independently by at least six contributors over roughly ten days and approximately 4,000 experiments, converged repeatedly on the same short list of conclusions about where its ceiling lies. Deep learning lost to tuned tree ensembles five times, including in this paper's new experiment, the first with a genuinely externally pretrained representation, which lost on real held-out data and rules out lack of external pretraining as the missing ingredient. Structural distance from the training distribution predicted error three times, with an unresolved target-dependent exception, and small-sample validation failed in three structurally distinct ways, each found by a different contributor. Six further interventions against one incumbent returned six characterized negative results rather than a single unexplained null. The convergence is the contribution: uncoordinated replication of a negative result is stronger evidence for a genuine ceiling than any single effort's result can be, and it deserves wider use wherever a problem draws sustained, semi-independent attention.

Data Availability and AI Disclosure

The dataset is proprietary to the ANRF AISEHack 2.0 Round 2 competition and is not publicly redistributable without host permission. The full experimental record (per-round code, event ledgers, candidate files, score records for all twelve of the authors' rounds and for §V-B) is retained and available for reviewer inspection on request, subject to the same permission. This program made extensive use of AI research assistants, multiple large language model agents operating both autonomously and under human direction, for experiment design, implementation, execution, and initial-draft synthesis. Every quantitative result reported here was independently verified against raw logs, ledgers, and score records by a human-directed review before inclusion, and none is reported solely on an agent's self-report. Adapt this disclosure to the target venue's policy before submission.

References

[1] R. A. Klein, K. A. Ratliff, M. Vianello, R. B. Adams, Š. Bahník, M. J. Bernstein *et al.*, "Investigating variation in replicability: A 'many labs' replication project," *Social Psychology*, vol. 45, no. 3, pp. 142–152, 2014.

[2] Open Science Collaboration, "Estimating the reproducibility of psychological science," *Science*, vol. 349, no. 6251, p. aac4716, 2015.

[3] R. Silberzahn, E. L. Uhlmann, D. P. Martin *et al.*, "Many analysts, one data set: Making transparent how variations in analytic choices affect results," *Advances in Methods and Practices in Psychological Science*, vol. 1, no. 3, pp. 337–356, 2018.

[4] J. Pineau, P. Vincent-Lamarre, K. Sinha, V. Larivière, A. Beygelzimer, F. d'Alché Buc, E. Fox, and H. Larochelle, "Improving reproducibility in machine learning research (a report from the NeurIPS 2019 reproducibility program)," *Journal of Machine Learning Research*, vol. 22, no. 164, pp. 1–20, 2021.
[5] X. Bouthillier, P. Delaunay, M. Bronzi, A. Trofimov, B. Nichyporuk, J. Szeto, N. Mohammadi Sepahvand, E. Raff, K. Madan, V. Voleti, S. Ebrahimi Kahou, V. Michalski, T. Arbel, C. Pal, G. Varoquaux, and P. Vincent, "Accounting for variance in machine learning benchmarks," in *Proceedings of Machine Learning and Systems (MLSys)*, 2021.
[6] S. Kapoor and A. Narayanan, "Leakage and the reproducibility crisis in machine-learning-based science," *Patterns*, vol. 4, no. 9, p. 100804, 2023.
[7] R. Schaeffer, J. Kazdan, Y. Denisov-Blanch, B. Miranda, M. Gerstgrasser, S. Zhang, A. Haupt, I. Gupta, E. Obbad, J. Dodge, J. Z. Forde, F. Orabona, S. Koyejo, and D. Donoho, "Position: Machine learning conferences should establish a 'refutations and critiques' track," *arXiv preprint arXiv:2506.19882*, 2025.
[8] F. Karl, L. M. Kemeter, G. Dax, and P. Sierak, "Position: Embracing negative results in machine learning," *arXiv preprint arXiv:2406.03980*, 2024.
[9] Ramprasad Research Group, "Machine-learning predictions of polymer properties with Polymer Genome," *Journal of Applied Physics*, vol. 128, no. 17, p. 171104, 2020.
[10] M. Aldeghi and C. W. Coley, "A graph representation of molecular ensembles for polymer property prediction," *Chemical Science*, vol. 13, pp. 10 486–10 498, 2022.
[11] C. Kuenneth and R. Ramprasad, "polyBERT: a chemical language model to enable fully machine-driven ultrafast polymer informatics," *Nature Communications*, vol. 14, 2023.
[12] L. Grinsztajn, E. Oyallon, and G. Varoquaux, "Why do tree-based models still outperform deep learning on tabular data?" in *Advances in Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track*, 2022.
[13] D. McElfresh, S. Khandagale, J. Valverde, V. Prasad C., G. Ramakrishnan, M. Goldblum, and C. White, "When do neural nets outperform boosted trees on tabular data?" in *Advances in Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track*, 2023.
[14] M. Praski, J. Adamczyk, and W. Czech, "Benchmarking pretrained molecular embedding models for molecular representation learning," *arXiv preprint arXiv:2508.06199*, 2025.
[15] K. K. Ben Hicham, J. G. Rittig, M. Grohe, and A. Mitsos, "Tabular foundation models for in-context prediction of molecular properties," *arXiv preprint arXiv:2604.16123*, 2026.
[16] S. Kang, "Multi-task learning for metal alloy property prediction: An empirical study of negative transfer and mitigation strategies," *arXiv preprint arXiv:2512.22740*, 2026.
[17] B. A. Eraqi, D. Khizbullin, S. S. Nagaraja, and S. M. Sarathy, "Molecular property prediction in the ultra-low data regime," *Communications Chemistry*, 2025.
[18] D. Horvath, G. Marcou, and A. Varnek, "Predicting the predictability: A unified approach to the applicability domain problem of QSAR models," *Journal of Chemical Information and Modeling*, vol. 49, no. 7, pp. 1762–1776, 2009.
[19] V. Agrawal, "Polymer property prediction using ensemble of directed message-passing neural networks with uncertainty estimation and applicability domain analysis," *MRS Communications*, 2026.
[20] R. Liu and A. Wallqvist, "Molecular similarity-based domain applicability metric efficiently identifies out-of-domain compounds," *Journal of Chemical Information and Modeling*, vol. 59, pp. 181–189, 2019, published online 2018-11-19.
[21] A. Gelman and E. Loken, "The statistical crisis in science: data-dependent analysis, a 'garden of forking paths,' explains why many statistically significant comparisons don't hold up," *American Scientist*, vol. 102, no. 6, pp. 460–465, 2014.
[22] D. Krstajic, L. J. Buturovic, D. E. Leahy, and S. Thomas, "Cross-validation pitfalls when selecting and assessing regression and classification models," *Journal of Cheminformatics*, vol. 6, p. 10, 2014.
[23] DeepChem, "ChemBERTa-77M-MTR," Hugging Face model repository, `DeepChem/ChemBERTa-77M-MTR`, 2022.
[24] C. Lu, C. Lu, R. T. Lange, J. Foerster, J. Clune, and D. Ha, "The AI scientist: Towards fully automated open-ended scientific discovery," arXiv preprint arXiv:2408.06292, 2024.